DECODE SIH 2026

OSCode by communityX

VoltGrid

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PROBLEM STATEMENT

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PROBLEM TITLE

Multimodal Deepfake & AI-Enabled Scam Detection Across Messages, Calls, Images & Video

WHO FACES THIS PROBLEM?

Every mobile user in India is a potential victim. Targets include seniors, rural populations, SMB owners, and students facing unknown phone calls, SMS, or deepfake video calls.

CURRENT GAPS IN THE SYSTEM

- Reactive, Not Real-Time: Apps identify known spam but cannot analyze live conversation content.
- Cloud Latency: Products that upload audio introduce 5–10 seconds of delay — enough time for scammers to extract an OTP.
- No Multimodal Coverage: No single pipeline fuses voice tone, video frames, text, and images simultaneously.
- Privacy Risks: Users hesitate to stream private calls to external cloud servers.

PROPOSED SOLUTION

WHAT ARE YOU BUILDING?

ScamShield is a privacy-first, real-time, multimodal scam & deepfake detection system that runs entirely on the user's device with zero network latency. It intercepts live audio calls, video calls, SMS, and image uploads, and analyses them through four specialised AI "Gates" TextGate, AudioGate, VideoGate, and ImageGate each producing a mathematical threat score from 0.0 to 1.0. If a threat is detected, the user is alerted within milliseconds before they can reveal sensitive information. For novel or complex threats, the system silently escalates (vectors only, never raw content) to a Cloud AI council of specialised fraud agents for a secondary verdict.

WHAT ARE THE 3-4 KEY FEATURES?

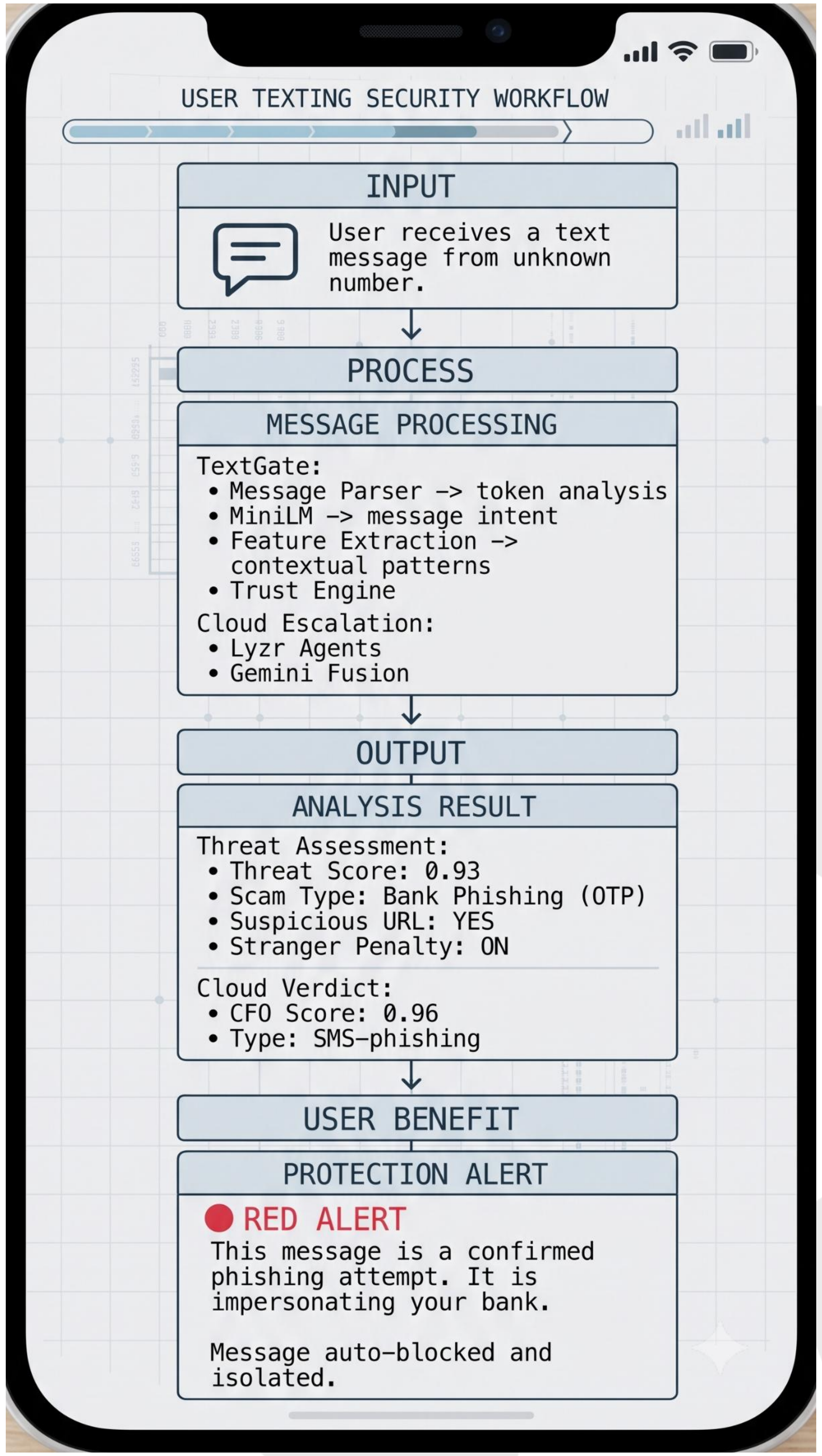
- **Zero-Latency Edge AI:** All four AI gates (text, audio, video, image) run locally on-device
- **Audio/Video Detection:**Analyzes audio frequencies and facial mechanics (like blinks and lip-syncing) to spot deepfakes and AI-generated voices.
- **Cloud Escalation:**Routes anonymized data from complex threats to specialized AI agents and live OSINT tools for advanced analysis.
- **Dynamic Trust Engine:**Adjusts security sensitivity based on the caller's reputation, applying stricter checks to unknown contacts while fast-tracking known ones.

WHAT MAKES IT DIFFERENT OR INNOVATIVE?

It is the only layer watching all four formats together as one continuous conversation. We utilize real, pretrained models and address a genuine research finding (the generalization gap where modern TTS beats 2019 baselines). We prioritize built-in explainability at every layer over a black-box score.

WORKING PROTOTYPE & DEMO

Functional Features
➢ Text Gate — SMS/chat scam detection (Sentence Transformers + keyword fallback)
➢ AudioGate — Live audio streaming via WebSocket + AI voice-clone detection (Librosa MFCC + Faster-Whisper)
➢ VideoGate — Deepfake detection via MediaPipe face mesh + audio track extraction
➢ ImageGate — OCR-based phishing detection (PyTesseract) + visual heuristics
➢ Cloud Escalation — Lyzr multi-agent council + Gemini fusion + Tavily OSINT
➢ Real-time React dashboard with WebSocket-driven threat alerts
➢ File upload analysis (text, audio, video, image)
➢ C2PA content authenticity verification
➢ SmolLM-135M local explainer (plain English threat explanations, offline)



TECH STACK & API/LIBRARIES

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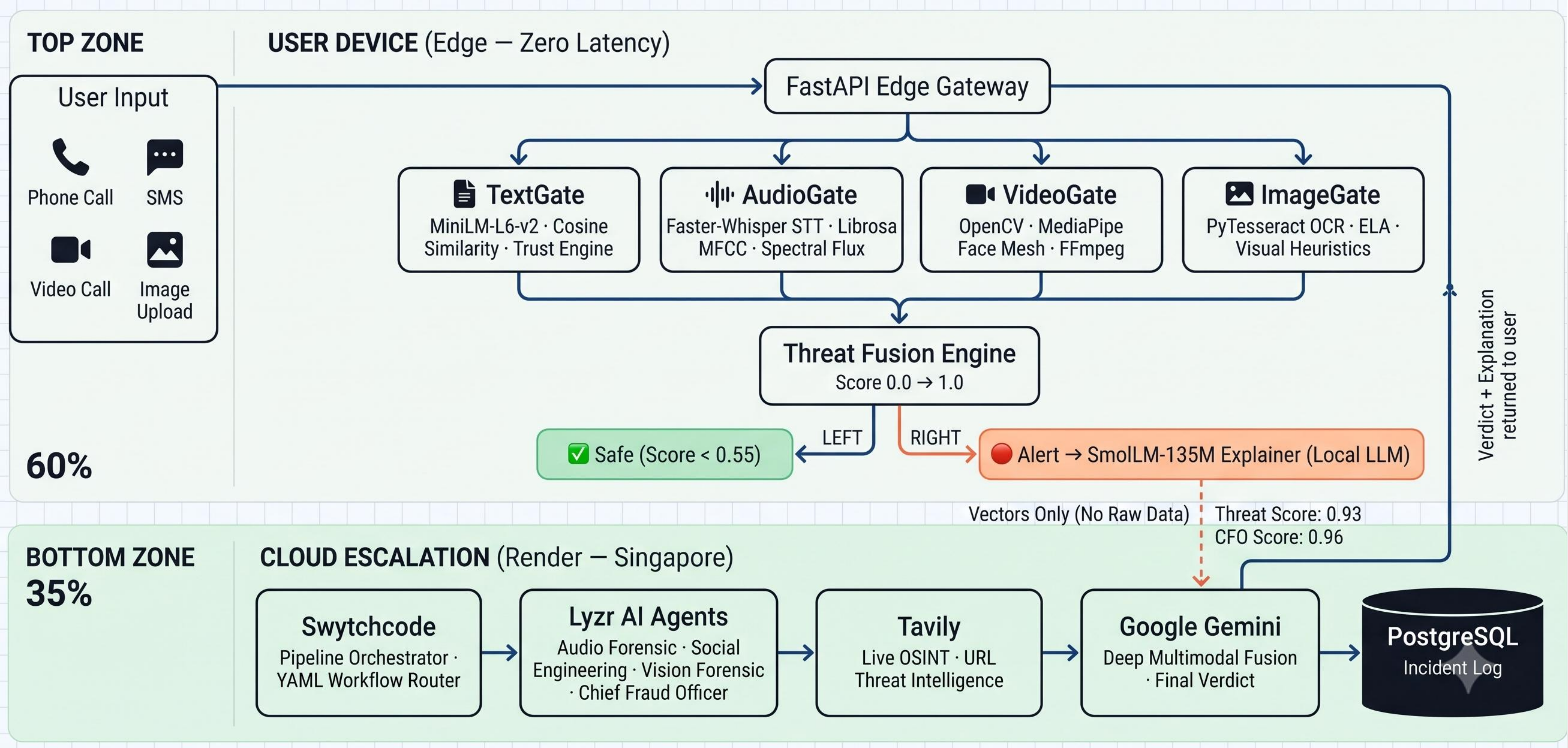
Layer	Technology
Frontend	React, Vite, TailwindCSS, Lucide Icons
Real-Time	WebRTC (simple-peer), Socket.io, MediaRecorder API
Backend (Edge)	Python 3, FastAPI, Uvicorn, Asyncio
AI/ML (Local)	Faster-Whisper (STT), SentenceTransformers (all-MiniLM-L6-v2), Librosa, SciPy, OpenCV, MediaPipe, PyTesseract, SmolLM-135M
Backend (Cloud)	FastAPI on Render (Singapore), PostgreSQL
Orchestration	Swytchcode Pipeline (YAML-defined workflow)
AI Partners	Lyzr AI (MCP agents), Google Gemini 3.6 Flash
Threat Intel	Tavily Search API
Database	SQLite (local), PostgreSQL (cloud)
Frontend	React, Vite, TailwindCSS, Lucide Icons

Category	Specific Libraries
NLP / Embeddings	sentence-transformers, all-MiniLM-L6-v2
Speech-to-Text	faster-whisper (CTranslate2 optimised)
Acoustic Analysis	librosa, scipy, numpy
Computer Vision	opencv-python, mediapipe, Pillow
OCR	pytesseract
Local LLM	SmolLM-135M-Instruct (HuggingFace)
Cloud AI	google-genai (Gemini), lyzr-agent-api, tavily-python
Web Framework	fastapi, uvicorn, httpx, pydantic
Category	Specific Libraries

ARCHITECTURE



PARTNERS

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LYZR AI

Powers a multi-agent fraud council that analyses scrubbed threat vectors in parallel and returns a consensus verdict with confidence score.

GOOGLE GEMINI

Provides deep multimodal reasoning for complex payloads that exceed local processing capacity, and synthesizes final human-readable explanations.

SWYTCHCODE

Orchestrates the entire cloud detection pipeline via a YAML-defined workflow — routing feature vectors to the correct agent based on modality

TAVILY

Delivers real-time OSINT by checking extracted URLs against global scam/phishing databases and returning threat evidence from the live web.

RENDER

Hosts the Cloud API (FastAPI) and managed PostgreSQL database,providing low-latency auto-scaling infrastructure and incident logging.

VALIDATION & FEASIBILITY

Test Category	What Was Tested	Results
Text Detection	50+ SMS samples: UPI fraud, OTP theft, lottery scams, bank phishing, safe messages	Scam texts consistently scored **> 0.75**; safe/family messages scored **< 0.25**. Zero false positives on verified contacts.
Audio Detection	AI-cloned voices (ElevenLabs, KikiVoice), real human conversations, police impersonation recordings	AI voice clones detected via low MFCC variance + spectral flux anomalies. Real voices passed acoustic tests. Semantic layer caught social engineering content independently.
Video Detection	Deepfake video samples + normal webcam recordings	MediaPipe face mesh detected frame jitter and blink anomalies in deepfakes. Clean videos passed with scores < 0.3.
Image Detection	Forged UPI payment screenshots, phishing WhatsApp messages, legitimate receipts	OCR + keyword matching flagged fake receipts. ELA (Error Level Analysis) detected pixel-level manipulation.
Cloud Escalation	End-to-end Lyzr agent council + Gemini fusion on all modalities	Multi-agent consensus improved accuracy on ambiguous cases. CFO correctly distinguished family members discussing fraud (safe) from impersonators committing fraud (scam).

Deployment Requirements

EDGE DEVICE REQUIREMENTS

- **CPU:** Any device with Python 3.8+ runtime (CPU only, no GPU needed).
- **RAM:** 2 GB (CPU-only).
- **Storage:** ~300 MB (one-time download) for AI Models (Whisper, LM, SmolLM).

CLOUD & BACKEND

- **Cloud:** Render free tier sufficient for pilot.
- **Database:** PostgreSQL for incident logging.
- **APIs:** Keys for Lyzr, Gemini, Tavily, Swytchcode.

INTEGRATION & SDK

- **Installation:** SDK via `pip install -e ..`
- **Embedded in:** Python apps, mobile wrappers, telecom middleware.

REGULATORY & COMPLIANCE

- **SMS Headers:** TRAI-compliant parsing (AD-SBIBNK, etc.).
- **Privacy:** No PII leaves device — GDPR/DPDPA compliant.

IMPACT

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RESPONSE TIME

< 100 ms on-device vs. 5–10 sec cloud-only solutions — 50× faster

COVERAGE

4 modalities (text + audio + video + image) in a single SDK — first of its kind

PRIVACY

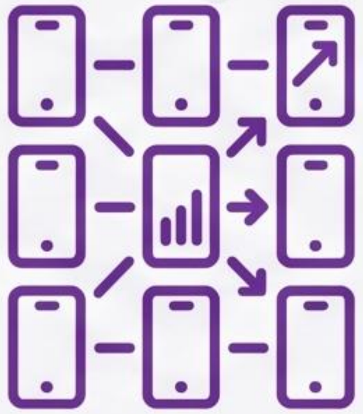
Zero raw data transmitted. 100% local processing for standard threats. Only mathematical vectors escalated.

COST

Edge-first design means 90%+ of detections need zero cloud compute — massively lower operating cost at scale

REACH

Works offline → protects users in rural areas with poor connectivity



SCALE

Each device runs independently → system scales linearly with users, not with server capacity

ROADMAP & CLOSING

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Phase 1 — COMPLETED —

Core Edge SDK

4 AI Gates: Text, Audio, Video, Image

- Local threat scoring · SmolLM-135M Explainer · SQLite Trust Engine · Offline-capable

Phase 2 — COMPLETED —

Cloud Escalation Pipeline

Multi-Partner Integration

- Lyzr AI Multi-Agent Council · Google Gemini Fusion · Tavily OSINT · Swytchoode Orchestration
- Render Deployment

Phase 3 — COMPLETED —

Real-Time Web Dashboard

React + WebSocket UI

- Live threat alerts · File upload analysis · WebRTC video/audio streaming · C2PA verification

Phase 4 — COMPLETED —

Mobile App (Limited)

Android Application

- Text Gate — functional · Audio Gate — Functional · Video & Image Gates — In development ·
- Core SDK integrated

Phase 5 — NEXT 3 MONTHS —

Telecom Partner Pilot

Carrier Integration

- Partner with 1–2 Indian telecom operators · Opt-in call screening service · TRAI-compliant SMS header parsing

Phase 6 — NEXT 6 MONTHS —

Federated Learning

Privacy-Preserving Model Updates

- Anonymised threat pattern sharing · Central model improvement
- Zero raw data transmission

Phase 7 — NEXT 12 MONTHS —

Multi-Language Support

Vernacular Scam Detection

- Hindi · Tamil · Telugu · Kannada
- Extend Whisper + MiniLM to Indian languages

4 of 7 Phases Complete

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PROJECT RESOURCES & DEMO

GitHub Repository

https://github.com/Hari-1925/ScamShield_SDK

(Includes Project Documentation, Architecture High-Res Diagrams)

Prototype Demo

https://drive.google.com/file/d/1IeCAcwMo5knx5loUB-Y4Z47Xw7BWb1Fm/view?usp=sharing

(Showcases the working prototype with all modalities validated)

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THANK YOU FOR THIS OPPUTUNITY